Applying the Daubert Standard to Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 01

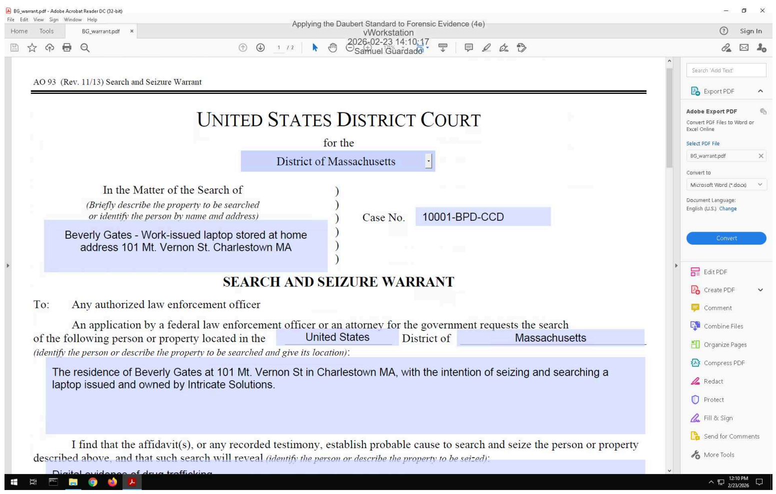
Student: Samuel Guardado
Email: sguar5@brockport.edu
Report Generated: Thursday, July 30, 2026 at 6:29 PM

Time on Task: 4 hours, 4 minutes
Progress: 100%

Section 1: Hands-On Demonstration

Part 1: Complete Chain of Custody Procedures

7. **Make a screen capture** showing the **contents of the search warrant in Adobe Reader**.



AO 93 (Rev. 11/13) Search and Seizure Warrant

UNITED STATES DISTRICT COURT
for the
District of Massachusetts

In the Matter of the Search of
(Briefly describe the property to be searched or identify the person by name and address)
Beverly Gates - Work-issued laptop stored at home address 101 Mt. Vernon St. Charlestown MA

Case No. 10001-BPD-CCD

SEARCH AND SEIZURE WARRANT

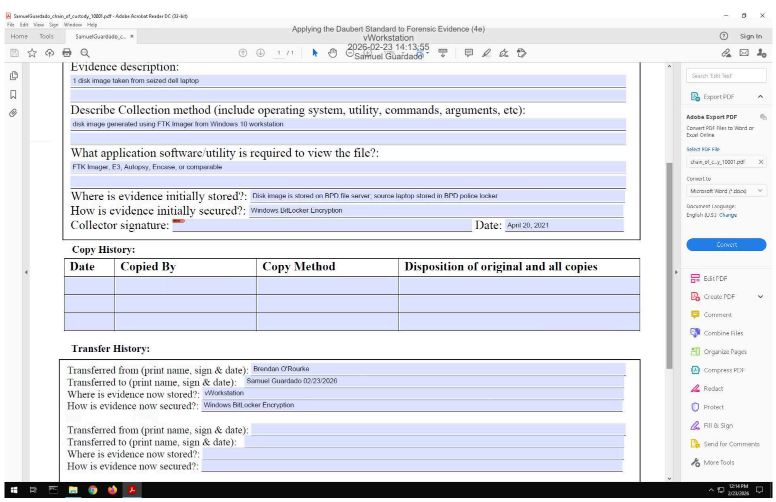
To: Any authorized law enforcement officer

An application by a federal law enforcement officer or an attorney for the government requests the search of the following person or property located in the United States District of Massachusetts
(identify the person or describe the property to be searched and give its location):

The residence of Beverly Gates at 101 Mt. Vernon St in Charlestown MA, with the intention of seizing and searching a laptop issued and owned by Intricate Solutions.

I find that the affidavit(s), or any recorded testimony, establish probable cause to search and seize the person or property described above, and that such search will reveal (identify the person or describe the property to be seized):

14. **Make a screen capture** showing the **completed Chain of Custody form in Adobe Reader.**



Evidence description:
1 disk image taken from seized dell laptop

Describe Collection method (include operating system, utility, commands, arguments, etc):
disk image generated using FTK Imager from Windows 10 workstation

What application software/utility is required to view the file?:
FTK Imager, E3, Autopsy, Encase, or comparable

Where is evidence initially stored?: Disk image is stored on BPD file server; source laptop stored in BPD police locker
How is evidence initially secured?: Windows BitLocker Encryption
Collector signature: Date: April 20, 2021

Copy History:

Date	Copied By	Copy Method	Disposition of original and all copies

Transfer History:

Transferred from (print name, sign & date): Brendan O'Rourke
Transferred to (print name, sign & date): Samuel Guardado 02/23/2026
Where is evidence now stored?: vWorkstation
How is evidence now secured?: Windows BitLocker Encryption

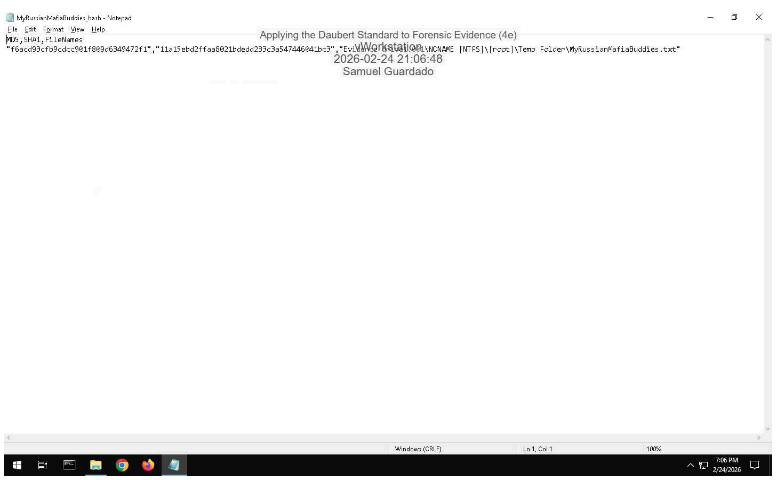
Transferred from (print name, sign & date):
Transferred to (print name, sign & date):
Where is evidence now stored?:
How is evidence now secured?:

Part 2: Extract Evidence Files and Create Hash Codes with FTK Imager

34. **Make a screen capture** showing the **contents of the 0002665_hash.csv file**.

37. **Make a screen capture** showing the **contents of the RecycleBinEvidence_hash.csv file**.

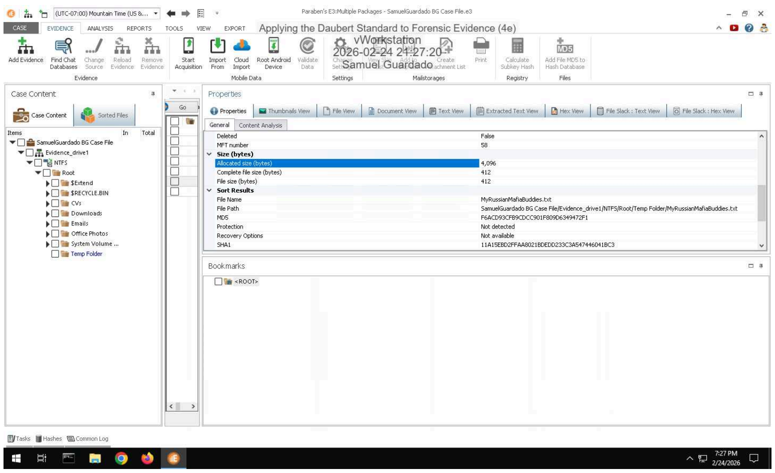
38. **Make a screen capture** showing the **contents of the MyRussianMafiaBuddies_hash.csv file**.



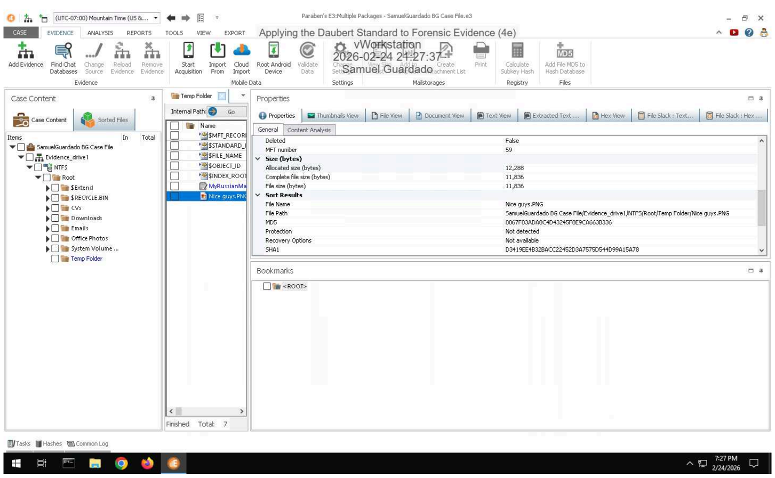
39. **Make a screen capture** showing the **contents of the Nice guys_hash.csv file**.

Part 3: Verify Hash Codes with E3

14. **Make a screen capture** showing the **MD5 and SHA1 values for the MyRussianMafiaBuddies.txt file**.



16. **Make a screen capture** showing the **MD5 and SHA1 values for the Nice Guys.png file**.



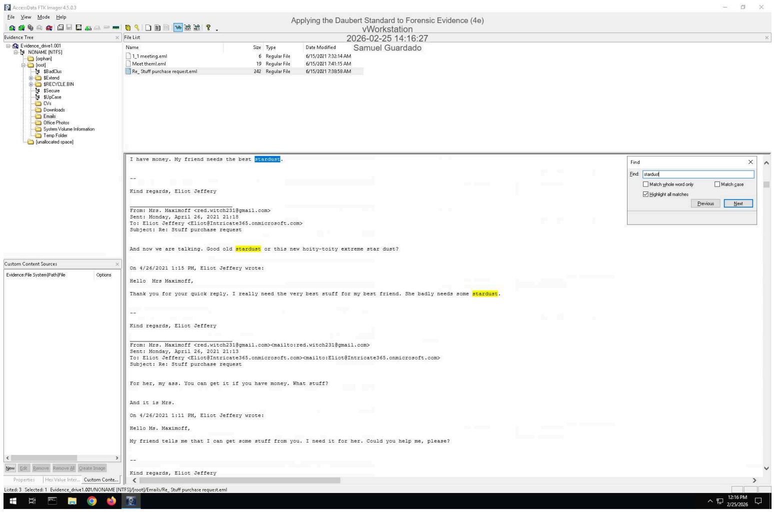
17. **Describe** how the hash values produced by E3 for the incriminating files compare to those produced by FTK. Do they match?

They do match. The only difference is that the one produced by FTK is in lowercase, and the one produced by EC3 is in uppercase.

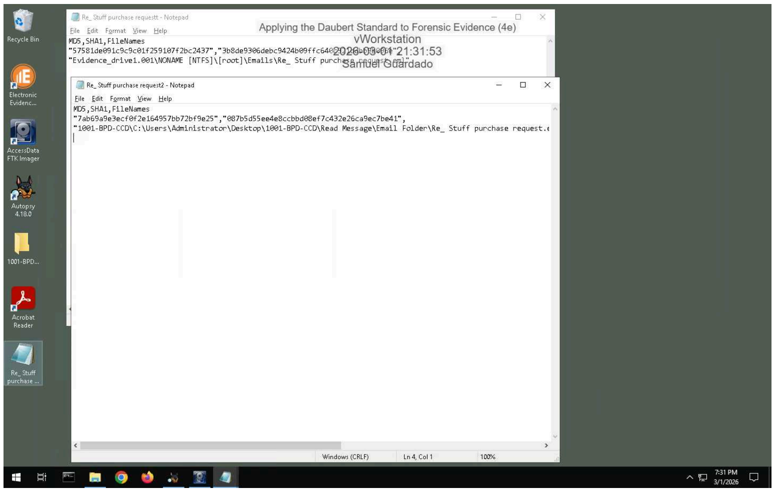
Section 2: Applied Learning

Part 1: Extract Evidence Files and Create Hash Codes with FTK Imager

5. **Make a screen capture** showing the **contents of the suspicious email file in the Display pane**.

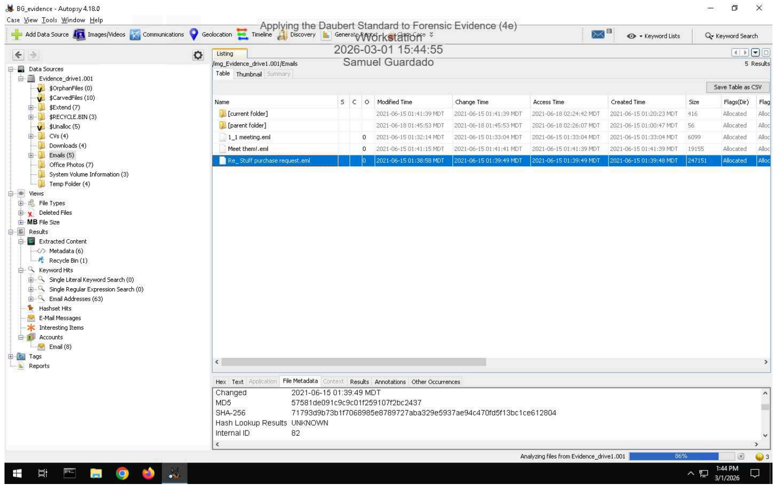


16. **Make a screen capture** showing the **two hash values for the suspicious email file**.



Part 2: Verify Hash Codes with Autopsy

11. **Make a screen capture** showing the **MD5 field in the Result Viewer**.

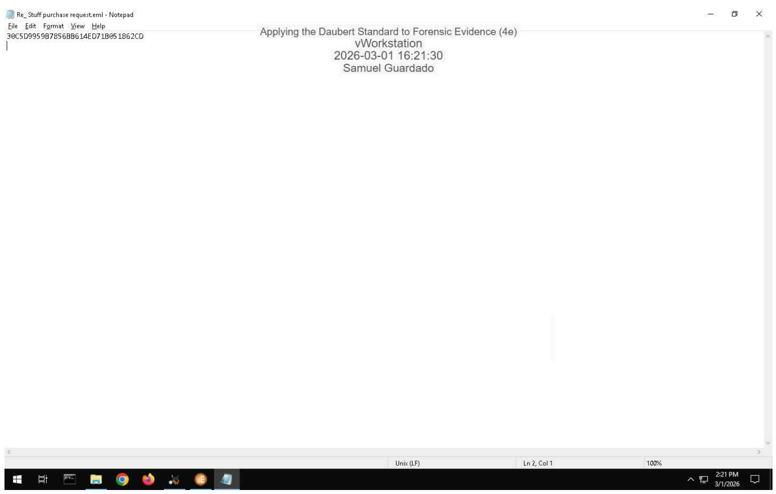


12. **Describe** how the hash value produced by Autopsy compares to the values produced by FTK Imager for the two .eml files.

The hash values of the incriminating files shown by Autopsy and FTK are identical, so we can be sure the files are authentic and have not been altered.

Part 3: Verify Hash Codes with E3

7. **Make a screen capture** showing the **MD5 value produced by E3**.



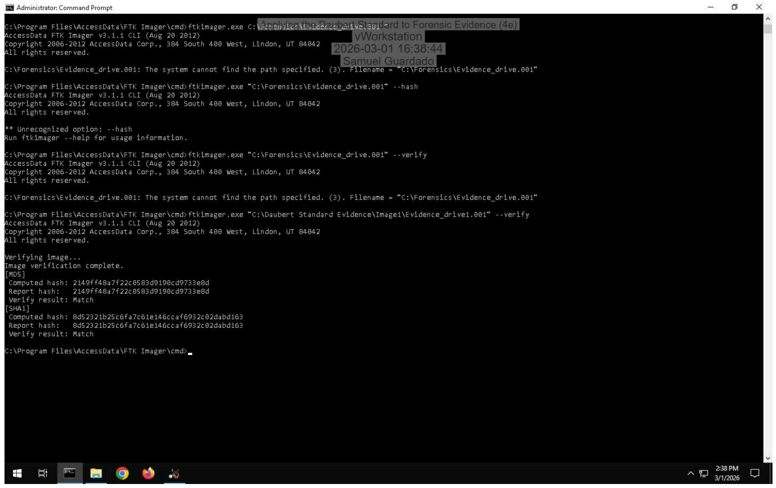
8. **Describe** how the hash value produced by E3 compares to the values produced by FTK Imager for the two .eml files and the value produced by Autopsy.

After examination of both of the incriminating pieces of evidence produced by FTK and E3, and Autopsy, we can confirm the evidence is valid because the hashvalues are the same.

Section 3: Challenge and Analysis

Part 1: Verify Hash Codes on the Command Line

Make a screen capture showing the **hash values for the Evidence_drive1.001 file**.



Part 2: Locate Additional Evidence

Define the original file names and file paths for each of the three files.

$R354ELH.xlsx original path and name was G:\VIP Info21DrugSales.xlsx \\ $RBQEEOTL/doc original path and name was G:\Students\manual-testing-fresher-resume-1.doc. \\ $RX3177E.pdf original path and name was G:\Work Doc\hr letter for visa.pdf.